

Bishop's University
Department of Computer Science
CS503 – Data Visualization
HW4 – Map and Network and Data Visualization

Problem: US-Covid19 Visualization and Analysis.

Hypothesis: There exists a correlation between the spatial distance among US states and the number of COVID-19 cases reported.

Objective: Validate the hypothesis visually using Map and Network data Analysis.

Data:

- Provided in a CSV file are COVID-19 cases recorded over a period in various US states.
- Python demos (references 1 and 3) illustrate the generation of a heatmap depicting COVID-19 cases, and also community detection.

Tasks: Data Analysis Using Adjacency Matrix:

- 1) Compute two metrics for each state: TNC (Total Number of Confirmed Cases) and TND (Total Number of Deaths).
- 2) Save the computed values in a CSV file with three columns: state, TNC, and TND.
- 3) Construct the Affinity Matrix A, where A_{ij} represents the similarity between state "i" and state "j" based on their TNC and TND.
 - Each state "i" is represented by a vector $x_i = [\text{TNC}, \text{TND}]$, enabling the computation of A_{ij} as follows:

$$A_{ij} = e^{-\|\frac{x_i - x_j}{\sigma}\|_2^2}$$

- Where σ is a free parameter set by the user.
- 4) Convert the affinity matrix A_{ij} to an Adjacency matrix B_{ij} by:
$$B_{ij} = 1 \quad \text{if } A_{ij} \geq 0.5$$
$$B_{ij} = 0 \quad \text{if } A_{ij} < 0.5$$
 - 5) Use the **NetworkX** Python library to generate the graph based on matrix B.
 - 6) Display the graph and label the nodes with the initials of the state names.
 - 7) Identify communities within the graph, where each community represents a cluster of similar states, and visualize these clusters.
 - 8) Change the value of σ and display the corresponding community detection results.
 - 9) Use Gephi software (reference 2) to address the aforementioned tasks, including displaying screenshots of the network and identified communities.
 - 10) Use the heatmap (reference 3) and the result of the community detection and try to determine if two state that are geographically close exhibit a similar number of COVID-19 cases. Offer a brief discussion on your findings.

References:

1. <https://www.kaggle.com/code/rahulgoel1106/community-detection-using-igraph-and-networkx>
2. <https://gephi.org/>
3. <https://bytescout.com/blog/plotting-geographical-heatmaps-using-python-folium-library.html>
4. <https://github.com/CambridgeUniversityPress/FirstCourseNetworkScience/blob/master/tutorials/Chapter%201%20Tutorial.ipynb>